COMPUTER PROGRAMMING
LAB 2

LAB TITLE

Basic Syntax and Input/Output in C++

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1. OBJECTIVE

The aim of this lab is to learn the basic structure of a C++ program. In this lab, we understand how to use input and output statements, work with various data types, follow proper indentation, and perform calculations using arithmetic operators.

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2. THEORY

A C++ program follows a particular structure which includes header files, the **main() function**, and program statements. The **iostream** library is commonly used to handle input and output operations.

The **cout** statement is used to display output on the screen, while **cin** is used to receive input from the user. The symbols << and >> are called stream insertion and stream extraction operators.

C++ is a statically typed programming language, meaning that each variable must be declared with a specific data type before it is used. Some common data types in C++ are:

- **int** – used for integer numbers
- **double** – used for decimal numbers
- **char** – used for single characters
- **bool** – used for true or false values
- **string** – used for text

C++ also provides arithmetic operators such as **+ , − , * , / , and %** which allow programmers to perform mathematical calculations.

Proper indentation improves the readability of programs. Comments are also important because they explain the code. In C++, comments can be written using // for single-line comments and / / for multi-line comments.

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3. SOFTWARE

- DEV-C++ IDE
- C++ Compiler (included with DEV-C++)

PROGRAM 1

Hello World and User Greeting

Algorithm

1. Start the program
2. Declare a variable to store the user's name
3. Ask the user to enter their name
4. Display a greeting message
5. End the program

```
#include <iostream>
using namespace std;

int main()
{
    string name;

    cout << "Enter your name: ";
    cin >> name;

    cout << "Hello " << name << "! Welcome to C++ programming.";

    return 0;
}
```

PROGRAM 2

Simple Calculator Using Arithmetic Operators

Algorithm

1. Start the program
2. Declare two variables for numbers
3. Take input values from the user
4. Perform addition, subtraction, multiplication, and division
5. Display the calculated results
6. End the program

```
#include <iostream>
using namespace std;

int main()
{
    int a, b;

    cout << "Enter two numbers: ";
    cin >> a >> b;

    cout << "Addition = " << a + b << endl;
    cout << "Subtraction = " << a - b << endl;
    cout << "Multiplication = " << a * b << endl;
    cout << "Division = " << a / b << endl;

    return 0;
}
```

Conclusion

In this lab, we learned about the basic structure of a C++ program. We practiced using input and output statements, worked with different data types, and performed calculations using arithmetic operators. This lab provided a basic understanding of C++ programming concepts.